

WELCOME

Hello Everyone I'm Marco

3rd year Mechatronics Engineering student, studying at AGH
University of Science and Thechnology, in Krakow, Poland.

1

Internship

7+

Months Experience

7+

Projects

Hire me →

Download CV



About Me



I am a Mechatronics Engineering student at AGH University of Science and Technology, with a strong focus on mechanical design and system integration.

I have hands-on experience in electronics, embedded systems, and prototyping, with particular strength in mechanical design and assembly.

Currently, I am working as an Engineering Intern at Laser MCR, where I conduct experimental testing on fiber and Nd:YAG laser systems, gaining experience with real engineering processes, materials, and parameter optimization.

I enjoy solving practical engineering problems where mechanical design, electronics, and control systems come together, applying these skills across academic and extracurricular projects.

I am particularly interested in robotics, research and development, and intelligent automation especially where theory meets real-world implementation. Currently, I am seeking internship or entry-level opportunities to contribute, learn, and grow in a professional engineering environment.

My Experience



Laser Systems Engineering Intern

Laser MCR | Kraków, Poland

Oct 2025 - Present

- ▶ Operated pulsed and CW fiber and Nd:YAG laser systems for material processing across metals, stone, and wood, in compliance with IEC 60825-1 laser safety standards.
- ▶ Optimized process parameters through iterative testing, eliminating laser-induced surface grid artifacts and achieving consistent, clean finishes.
- ▶ Diagnosed and resolved laser-material interaction issues using systematic troubleshooting and parameter isolation techniques.
- ▶ Analyzed 250+ experimental trials using Excel to identify performance trends and support data-driven optimization.
- ▶ Produced structured technical reports and maintained detailed engineering documentation aligned with laboratory standards.
- ▶ Supported hardware configuration and system setup through component selection using technical datasheets and supplier catalogues.
- ▶ Designed 3D models and developed custom 3D-printed components to support experimental setup efficiency and system integration.

Visit Laser MCR [↗](#)

Latest Project

All images are from real projects, enhanced with AI to improve visual quality for web presentation.

Thesis Project: Upcoming

Vision-Based Tracking and Motion Prediction of Objects on a Pan-Tilt Mechatronic System

[Vision Tracking & Prediction](#)[Pan-Tilt](#)[More info →](#)

Celestial Navigation System

An autonomous sun & star tracking navigation device using sensors and microcontrollers for position determination.

[Vision Systems](#)[Raspberry Pi](#)[More info →](#)

5-DOF Robotic Manipulator

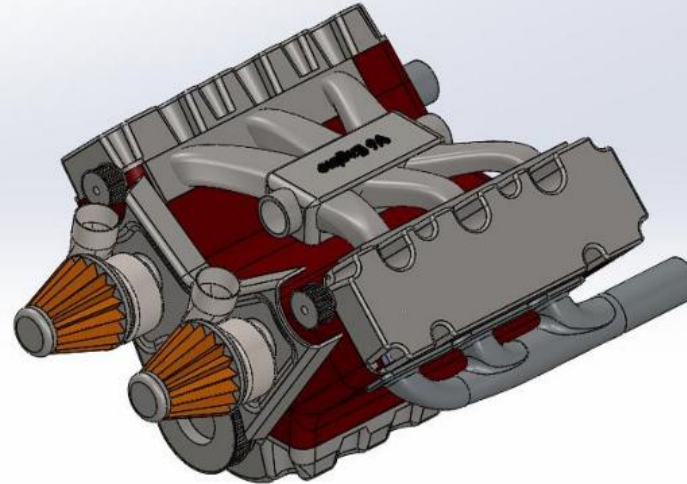
A five degree-of-freedom robotic arm with inverse kinematics control and precision gripping.

[Servos](#)[Arduino](#)[3D Print](#)[Mor](#) [Edit with Lovable](#) [×](#)

[Vision Tracking & Prediction](#)
[Pan-tilt](#)
[More info →](#)
[Vision Systems](#)
[Raspberry Pi](#)
[More info →](#)
[Servos](#)
[Arduino](#)
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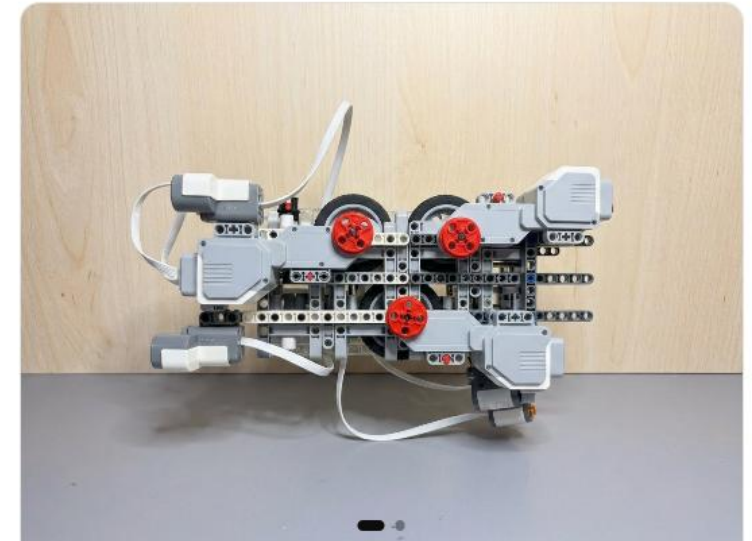

Voice-Activated Smart Door

A smart door system using voice recognition to authenticate and grant access to authorised users.

[Signal Processing & Identification](#)
[Arduino](#)
[More info →](#)


V6 Engine

A detailed CAD model and simulation of a V6 internal combustion engine with full assembly.

[SolidWorks](#)
[Siemens NX](#)
[More info →](#)


Rope Climbing Robot

A compact mechatronic robot designed to autonomously climb ropes using friction-driven wheels.

[Husarion](#)
[Motors](#)
[More info →](#)

Technical Skills



Mechanical & Design

-  2D Technical Drawings
-  3D Modelling & Assembly
-  GD&T (Fundamentals)
-  Basic FEM Analysis
-  Motion & Kinematic Analysis

Software



AutoCAD



SolidWorks



Siemens NX



Electronics & Embedded

-  Actuator & Sensor Integration
-  Arduino (Servo, PWM, PID)
-  ESP32 (Embedded Applications)
-  Raspberry Pi (Vision Systems)



Programming & Software

-  C++ (OOPSE)
-  Python (Scripting)
-  MATLAB (Control & Signals)
-  Simulink (Modeling & Sim)

Software

C++

C++



Python



MATLAB



Simulink

My Certificates



MACQUARIE
University
SYDNEY · AUSTRALIA

Aug 24, 2025

Aung Pyae Sone

has successfully completed

Excel Skills for Business: Advanced

an online course authorized by Macquarie University and offered through Coursera



Nicky Bell, Lecturer and Lead Course Designer
Dr Prabhu S.M. Karimnasser, Lecturer Teaching &
Leadership (Department of Actuarial Studies & Business
Analysis)
Associate Professor, Director, Learning & Teaching
Innovation Centre

Verify at:
<https://coursera.org/verify/NF92HCUT5uPA>

Coursera has confirmed the identity of this individual and their participation in the course.

This certificate attests to the learner's completion of an online course / project delivered via Coursera. It does not constitute formal endorsement or accreditation or entry and does not itself grant academic credit, grades or a degree. Institutions or employers may, at their discretion, recognize this learning toward their own programs or credentials.

Excel Skills for Business: Advanced

Macquarie University · Aug 2025



Microsoft

May 4, 2025

Aung Pyae Sone

has successfully completed

Power BI & Power Virtual Agents

an online course authorized by Microsoft and offered through Coursera



Marni Baker-Smith,
Chief Content Officer

Verify at:
<https://coursera.org/verify/v28T1Q27056u6>

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Power BI & Power Virtual Agents

Microsoft · May 2025

Get In Touch

I'm currently seeking internship or entry-level opportunities. Feel free to reach out via LinkedIn!

 [Connect on LinkedIn](#)